Where Z is the atomic number and 'S' is the screening constant which can be calculated using Slater's rules as described below.

Step 1 :

Write the electronic configuration of the atom and rearrange it by grouping ns and np orbitals together and others separately in the following form.

(1s) (2s, 2p) (3s, 3p) (3d) (4s, 4p) (4d) (4f) (5s, 5p)...

Step 2 :

Identify the group in which the electron of interest is present. The electron present right to this group does not contribute to the shielding effect.

Each of the electrons within the identified group (denoted by 'n') shields to an extent of 0.35 unit of nuclear charge. However, it is 0.30 unit for 1s electron.

Step 3 :

Shielding of inner shell electrons.

If the electron of interest belongs to either s or p orbital,

i) each electron within the (n-1) group shields to an extent of 0.85 unit of nuclear charge, and

ii) each electron within the (n-2) group (or) even lesser group (n-3), (n-4) etc... completely shields i.e. to an extent of 1.00 unit of nuclear charge.

If the electron of interest belongs to d or f orbital, then each of electron left of the group of electron of interest shields to an extent of 1.00 unit of nuclear charge.

Step 4 :

Summation of the shielding effect of all the electrons gives the shielding constant 'S'

Example: Let us explain the calculation of effective nuclear charge on 4s electron and 3d electron in scandium. The electronic configuration of scandium is $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $4s^2$, $3d^1$. we can rearrange as below.

$\underbrace{(1s)^2}_{(n-3)}\ \underbrace{(2s,2p)^8}_{(n-2)}\ \underbrace{(3s,3p)^8\,(3d)^1}_{(n-1)}\ \underbrace{(4s)^2}_{n}$

Group	number of electron in the group	contribution of each electron to 'S' value	contribution of a particular group to electrons to 'S' value
(n)	1	0.35	0.35
(n-1)	9	0.85	7.65
(n-2) & others	10	1	10.00
S value			18.00

$Z_{eff} = Z - S$ i.e.= 21- 18 $\therefore Z_{eff} = 3$

Calculation of effective nuclear charge on 3d electron

$\underbrace{(1s)^2}_{(n-3)}\ \underbrace{(2s,2p)^8}_{(n-2)}\ \underbrace{(3s,3p)^8}_{(n-1)}\ (3d)^1\ \underbrace{(4s)^2}_{n}$

Group	number of electron in the group	contribution of each electron to 'S' value	contribution of a particular group to electrons to 'S' value
n	0	0.35	0
(n-1) & others	18	1	18
S Value			18

$\therefore Z_{eff} = Z - S$ i.e. =21 - 18 $\therefore Z_{eff} = 3$

11th Std Chemistry 068-099.indd 81 23-12-2021 15:10:14